

## JACKSON NWS, KY, USA

WMO: 724190

Lat: 37.5911N

Lon: 83.3139W

Elev: 1358

StdP: 13.99

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03889

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.7        | 15.1 | -1.6                          | 5.3 | 14.7 | 3.4 | 6.9 | 18.2 | 17.6                  | 39.3 | 15.9 | 39.6 | 4.2                   | 240  | 0.356 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.3                   | 89.5            | 73.5 | 87.3 | 72.8 | 85.1 | 72.0 | 76.6                | 85.4 | 75.3 | 83.4 | 74.1 | 81.9 | 3.3                  | 290  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.8                            | 132.5 | 81.8 | 72.7 | 127.5 | 80.2 | 71.7 | 123.2 | 78.5 | 40.9          | 85.4 | 39.6 | 83.3 | 38.5 | 82.0 | 82.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |       |    |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|-------|----|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       | DB    | WB |
| Min               | Max  | Min  | Max                        | Min  | Max  | Min                | Max                                                | Min  | Max        | Min  | Max        | Min   | Max        | Min   |       |    |
| (a)               | (b)  | (c)  | (d)                        | (e)  | (f)  | (g)                | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)   | (n)        | (o)   |       |    |
| 13.9              | 11.8 | 10.4 |                            | 3.2  | 93.0 | 6.9                | 3.7                                                | -1.8 | 95.7       | -5.8 | 97.8       | -9.7  | 99.9       | -14.7 | 102.6 |    |
|                   |      |      |                            | 2.2  | 78.9 | 6.6                | 1.4                                                | -2.6 | 79.9       | -6.5 | 80.7       | -10.2 | 81.5       | -15.0 | 82.6  |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 56.9   | 35.1  | 39.2  | 47.3  | 58.1  | 65.2  | 72.2  | 75.0  | 74.6  | 68.8  | 58.4  | 47.9  | 39.5  |
|                                                                        | DBStd       | 16.68  | 12.68 | 11.85 | 11.52 | 9.82  | 7.61  | 5.06  | 4.11  | 4.41  | 6.74  | 8.79  | 10.03 | 10.80 |
|                                                                        | HDD50       | 1593   | 486   | 336   | 192   | 37    | 3     | 0     | 0     | 0     | 0     | 26    | 157   | 356   |
|                                                                        | HDD65       | 4142   | 926   | 723   | 555   | 247   | 95    | 9     | 1     | 1     | 39    | 239   | 518   | 789   |
|                                                                        | CDD50       | 4101   | 25    | 34    | 109   | 279   | 475   | 667   | 774   | 762   | 565   | 286   | 94    | 31    |
|                                                                        | CDD65       | 1176   | 0     | 1     | 7     | 39    | 103   | 226   | 310   | 298   | 154   | 34    | 4     | 0     |
|                                                                        | CDH74       | 8103   | 0     | 3     | 46    | 297   | 677   | 1522  | 2181  | 2071  | 1078  | 213   | 15    | 0     |
|                                                                        | CDH80       | 2293   | 0     | 0     | 3     | 48    | 136   | 415   | 671   | 657   | 328   | 35    | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 4.0    | 5.4   | 5.4   | 5.5   | 5.3   | 3.5   | 2.7   | 2.3   | 2.3   | 2.5   | 3.4   | 4.5   | 4.9   |
| (15) Precipitation                                                     | PrecAvg     | 47.4   | 3.0   | 4.0   | 4.0   | 3.5   | 5.2   | 4.9   | 4.8   | 3.9   | 3.3   | 3.1   | 4.1   | 4.4   |
|                                                                        | PrecMax     | 63.3   | 6.5   | 7.9   | 9.8   | 10.0  | 10.8  | 9.1   | 9.7   | 7.7   | 7.8   | 7.4   | 9.3   | 13.0  |
|                                                                        | PrecMin     | 34.4   | 0.8   | 1.3   | 1.5   | 0.8   | 2.2   | 1.4   | 2.2   | 1.5   | 1.1   | 0.5   | 1.5   | 1.7   |
|                                                                        | PrecStd     | 8.0    | 1.3   | 1.8   | 2.2   | 2.0   | 1.9   | 2.2   | 1.9   | 1.7   | 2.0   | 1.8   | 2.0   | 2.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 67.8  | 71.7  | 79.1  | 84.5  | 86.6  | 89.7  | 91.8  | 93.1  | 91.2  | 83.9  | 75.9  | 68.4  |
|                                                                        |             | MCWB   | 57.6  | 58.5  | 60.5  | 63.0  | 70.3  | 73.6  | 76.0  | 73.6  | 69.3  | 67.3  | 61.8  | 58.4  |
|                                                                        | 2%          | DB     | 62.8  | 66.2  | 74.0  | 80.9  | 83.6  | 87.0  | 88.6  | 89.1  | 87.2  | 79.3  | 70.9  | 63.4  |
|                                                                        |             | MCWB   | 55.1  | 54.8  | 57.7  | 61.3  | 68.0  | 73.1  | 75.0  | 73.7  | 70.0  | 65.2  | 57.7  | 55.4  |
|                                                                        | 5%          | DB     | 58.7  | 61.2  | 69.5  | 77.5  | 81.1  | 84.5  | 86.3  | 86.3  | 83.9  | 75.8  | 66.6  | 59.6  |
|                                                                        |             | MCWB   | 53.0  | 51.9  | 55.4  | 60.2  | 67.2  | 72.0  | 74.2  | 73.2  | 69.5  | 63.0  | 55.8  | 53.0  |
|                                                                        | 10%         | DB     | 54.2  | 56.6  | 64.8  | 73.7  | 78.2  | 82.1  | 83.9  | 83.7  | 80.4  | 72.1  | 63.0  | 56.0  |
|                                                                        |             | MCWB   | 48.2  | 49.2  | 53.0  | 58.3  | 66.0  | 70.6  | 73.1  | 72.2  | 67.8  | 61.4  | 53.8  | 50.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 59.8  | 61.2  | 62.2  | 67.1  | 73.2  | 77.0  | 78.4  | 77.8  | 74.5  | 71.5  | 64.1  | 60.7  |
|                                                                        |             | MCDB   | 64.8  | 68.5  | 74.7  | 78.7  | 82.8  | 85.9  | 87.6  | 87.6  | 84.5  | 79.6  | 72.8  | 65.9  |
|                                                                        | 2%          | WB     | 56.9  | 57.5  | 60.0  | 64.6  | 71.1  | 75.0  | 77.1  | 76.2  | 72.8  | 68.4  | 61.0  | 57.5  |
|                                                                        |             | MCDB   | 61.9  | 63.4  | 70.5  | 75.7  | 79.9  | 83.7  | 85.8  | 85.3  | 82.5  | 74.6  | 67.9  | 61.4  |
|                                                                        | 5%          | WB     | 53.7  | 54.0  | 57.7  | 62.4  | 69.2  | 73.7  | 75.7  | 74.8  | 71.1  | 65.8  | 58.1  | 54.6  |
|                                                                        |             | MCDB   | 58.2  | 58.8  | 66.6  | 72.4  | 77.7  | 82.0  | 83.3  | 83.0  | 79.5  | 73.1  | 64.4  | 58.4  |
|                                                                        | 10%         | WB     | 49.2  | 50.3  | 54.9  | 60.5  | 67.6  | 72.2  | 74.4  | 73.5  | 69.7  | 63.4  | 54.9  | 51.0  |
|                                                                        |             | MCDB   | 53.7  | 56.0  | 62.7  | 69.7  | 75.5  | 79.8  | 81.5  | 80.9  | 76.7  | 69.7  | 60.8  | 55.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.3  | 16.5  | 18.6  | 19.8  | 18.0  | 17.2  | 16.3  | 17.0  | 17.9  | 18.3  | 16.8  | 14.4  |
|                                                                        |             | MCDBR  | 18.2  | 22.2  | 22.8  | 23.2  | 20.5  | 19.6  | 19.0  | 19.9  | 21.7  | 22.0  | 20.0  | 18.5  |
|                                                                        |             | MCWBR  | 12.5  | 15.1  | 12.6  | 11.6  | 9.8   | 9.2   | 8.3   | 8.0   | 8.4   | 10.4  | 11.5  | 13.7  |
|                                                                        | 5% WB       | MCDBR  | 16.8  | 20.0  | 19.8  | 19.4  | 17.7  | 17.4  | 16.7  | 17.0  | 17.8  | 18.0  | 16.6  | 16.8  |
|                                                                        |             | MCWBR  | 12.8  | 15.9  | 12.6  | 11.0  | 9.3   | 9.1   | 8.4   | 8.0   | 8.0   | 9.9   | 12.0  | 13.9  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.300 | 0.312 | 0.337 | 0.382 | 0.427 | 0.444 | 0.471 | 0.478 | 0.396 | 0.351 | 0.321 | 0.297 |
|                                                                        | taud        |        | 2.520 | 2.478 | 2.450 | 2.326 | 2.208 | 2.201 | 2.148 | 2.115 | 2.329 | 2.459 | 2.513 | 2.542 |
|                                                                        | Ebn at Noon |        | 283   | 293   | 294   | 285   | 272   | 267   | 258   | 253   | 270   | 274   | 272   | 277   |
|                                                                        | Edh at Noon |        | 25    | 29    | 33    | 40    | 45    | 46    | 48    | 49    | 37    | 30    | 25    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 618   | 821   | 1177  | 1555  | 1746  | 1936  | 1806  | 1705  | 1424  | 1076  | 765   | 543   |
|                                                                        | RadStd      |        | 66    | 117   | 102   | 116   | 159   | 108   | 103   | 94    | 146   | 138   | 87    | 58    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +200  | +119  |
| (47) Regional (22 neighbors) | +0.77 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +227  | +128  |

Nomenclature: See separate page